FINITE GENERATION OF THE TOP DIRECT IMAGE FOR PROPER MORPHISMS WITH BOUNDED FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Noetherian finiteness of proper cohomology. Cohomology of Schemes, Lemma coherent-lemma-proper-over-affine-cohomology-finite

Source reference. Original reference 01XJ: sources/stacks-chow/tags.txt, source lines 2520–2520. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09ZR: sources/stacks-chow/tags.txt, source lines 12609–12609. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05M5: sources/stacks-chow/tags.txt, source lines 7259–7259. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01F6: sources/stacks-chow/tags.txt, source lines 1859–1859. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 086M: sources/stacks-chow/tags.txt, source lines 10397–10397. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0E7D: sources/stacks-chow/tags.txt, source lines 17778–17778. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZR: sources/stacks-chow/tags.txt, source lines 2604–2604. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0EX3: sources/stacks-chow/tags.txt, source lines 18643–18643. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 02O6: sources/stacks-chow/tags.txt, source lines 3505–3505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $d \geq 0$. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Assume*

(1) *f is a proper morphism all of whose fibres have dimension $\leq d$,*
(2) *$\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module of finite type.*

Then $R^d f_* \mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module of finite type.

Proof. The module $R^d f_* \mathcal{F}$ is quasi-coherent by Cohomology of Schemes, Lemma 01XJ. The question is local on Y hence we may assume Y is affine. Say $Y = \mathrm{Spec}(R)$. Then it suffices to prove that $H^d(X, \mathcal{F})$ is a finite R-module.

By Lemma 09ZR we can write $X = \lim X_i$ as a cofiltered limit with $X_i \to Y$ proper and of finite presentation and such that both $X \to X_i$ and transition morphisms are closed immersions. For some i we have that $X_i \to Y$ has fibres of dimension $\leq d$, see Lemma 05M5. We have $R^p f_* \mathcal{F} = R^p f_{i,*}(X \to X_i)_* \mathcal{F}$ by Cohomology of Schemes, Lemma 01XC and Leray (Cohomology, Lemma 01F6). Thus we may replace X by X_i and reduce to the case discussed in the next paragraph.

Assume Y is affine and $f : X \to Y$ is proper and of finite presentation and all fibres have dimension $\leq d$. We can write $\mathcal{F}$ as a quotient of a finitely presented $\mathcal{O}_X$-module $\mathcal{F}'$, see Properties, Lemma 086M. The map $H^d(X, \mathcal{F}') \to H^d(X, \mathcal{F})$ is surjective, as we have $H^{d+1}(X, \mathrm{Ker}(\mathcal{F}' \to \mathcal{F})) = 0$ by the vanishing of higher cohomology seen in Lemma 0E7D (or its proof). Thus we reduce to the case discussed in the next paragraph.

Assume $Y = \mathrm{Spec}(R)$ is affine and $f : X \to Y$ is proper and of finite presentation and all fibres have dimension $\leq d$ and $\mathcal{F}$ is an $\mathcal{O}_X$-module of finite presentation. Write $Y = \lim_{i \in I} Y_i$ as a cofiltered limit of affine schemes with $Y_i = \mathrm{Spec}(R_i)$ the spectrum of a Noetherian ring (for example a finite type $\mathbf{Z}$-algebra). We can choose an element $0 \in I$ and a finite type morphism $X_0 \to Y_0$ such that $X \cong Y \times_{Y_0} X_0$, see Lemma 01ZM. After increasing 0 we may assume $X_0 \to Y_0$ is proper (Lemma 081F) and that the fibres of $X_0 \to Y_0$ have dimension $\leq d$ (Lemma 05M5). After increasing 0 we can assume there is a coherent $\mathcal{O}_{X_0}$-module $\mathcal{F}_0$ which pulls back to $\mathcal{F}$, see Lemma 01ZR. By Lemma 0EX3 we have

$$H^d(X, \mathcal{F}) = H^d(X_0, \mathcal{F}_0) \otimes_{R_0} R$$

This finishes the proof because the cohomology module $H^d(X_0, \mathcal{F}_0)$ is finite by Cohomology of Schemes, Lemma 02O6. □